Semester 3

Course Code	19U3CRCMT4
Title of the course	Vector Calculus, Trigonometry and Matrices
Semester in which the course is to be taught	3
No. of credits	4
No. of contact hours per week	5
Total Hours	90

Text Book:

1. Engineering Mathematics, N.P. Bali, Manish Goyal

2.S.L. Loney – Plane Trigonometry Part – II, S. Chand and Company Ltd.

Module 1 **(20 hrs)**

Scalar and Vector Fields, Gradient of a Scalar Field, Geometrical Interpretation of Gradient, Directional Derivative, Properties of Gradient, Divergence of a Vector Point Function, Curl of a Vector Point Function, Physical Interpretation of Divergence, Physical Interpretation of Curl, Properties of Divergence and Curl, Repeated Operations by ∇ .

(Sections 8.10-8.20 of Text 1)

Module 2 **(25 hrs)**

Integration of Vector Functions, Line Integrals, Circulation, Work Done by a Force, Surface Integrals, Volume Integrals, Divergence Theorem of Gauss (Relation between Surface and Volume Integrals), Green's Theorem in the Plane, Stoke's Theorem (Relation between Line and Surface Integrals).(All theorems without proof).

(Sections 8.21-8.29 of Text 1)

Module 3 **(20 hrs)**

Trigonometry

Circular and Hyperbolic functions of complex variables, Separation of functions of complex variables into real and imaginary parts, Factorization of x^n+1, x^n-1, x^{2n}-$2x^n a^n cos(n\theta)$+a^{2n} and summation of infinite series by C+iS method.

(Relevant sections of Text 2- Chapter V,VI,VIII,IX)

Module IV **(25 hrs)**

Matrices:

Elementary Transformations, Elementary Matrices, Inverse of Matrix by E-operations (Gauss-Jordan method), Rank of a Matrix, Solution of a System of Linear Equations, If A is a Non-Singular Matrix, then the Matrix Equation AX = B has a Unique Solution, Vectors, Linear dependence and Linear Independence of Vectors, Linear Transformations, Orthogonal Transformation, Complex Matrices, Characteristic Equation, Eigen Vectors, Cayley Hamilton Theorem

(Sections 3.34-3.35 and 3.37-3.48 of Text 1)

References

1.Erwin Kreyszig : Advanced Engineering Mathematics, 8th ed., Wiley.

2. H.F. Davis and A.D. Snider: Introduction to Vector Analysis, 6th ed., Universal Book Stall, New Delhi.

3.Shanti Narayan, P.K Mittal – Vector Calculus (S. Chand)

4.Merle C. Potter, J. L. Goldberg, E. F. Aboufadel – Advanced Engineering Mathematics (Oxford)
5.Ghosh, Maity – Vector Analysis (New Central books)
6.Shanti Narayan - Matrices (S. Chand & Company)

Question Paper Pattern

MODULE	Part A (2 Marks) Answer any 10	Part B (5 Marks) Answer any 5	Part C (10 Marks) Answer any 3
1	4	2	1
2	3	2	1
3	2	2	1
4	3	2	1
TOTAL	12	8	4